AI-Use Disclosure

IBM Qiskit Fall Fest 2026 | QC@IITI, IIT Indore | Industry Challenge by QryptoNox

Mandatory 1-Page Disclosure Form [Rules Section 10, 12, 14]

1. Tools and Environment

Item	Specification & Version
AI Assistant Used	Antigravity AI Coding Assistant (Gemini 3.8 Flash model)
Submission Libraries	Qiskit 2.5.2, NumPy 2.5.4, SciPy 1.18.1, Python Standard Library
Testing & Dev Environment	Pytest 9.1.1, Colorama 0.4.6 (isolated in requirements-dev.txt and local/)

2. What AI Assistance Was Used For

1. Test-Driven Development (TDD) Workflow: Writing comprehensive test suites (tests/) before solver implementation.

2. Qiskit 2.x Circuit Construction: Synthesis of Gate('oracle', n+m, []) placeholder circuits, little-endian bit decoders, and Fast Walsh-Hadamard Transform in O(N log N).

3. Local Simulation Engine: Designing a vectorized basis-state permutation runner for 16-qubit oracles that eliminated 64 GiB matrix allocation, running in 1 MB RAM and 0.0004s.

3. Independent Human Verification & What AI Got Wrong

Every formula and algorithm proposed by the AI was independently checked against empirical quantum simulations:

Hypothesis / Component	AI Claim [HAND-CALC]	Independent Simulation Result	Action Taken
Grover Recovery (E-02)	Claimed k=4..5 iterations in 2 shots	Verified >85% probability success empirically at k=4 for n=14, allocated 6 shots (2x buffer)	Adopted honest fa
Walsh Threshold at Stress Noise (E-05)*	Stated 2 shots* shots threshold assumed ideal	Verified 6 shots needed at Stress Noise p=0.12, Standard 2 shots fail	**Replaced static pu with Wilson-statistics gate (gap <= 10) and XO**
Qiskit Endianness (Q6)	Hand-coded CNOT matrix inverted Qiskit's little-endian bit ordering	Caught and corrected immediately (got 01, expected 11)	Adopted Qiskit product order to match Qiskit convention.
BV Readout Fidelity (E-01)	Derived p(0^8) = (1 - p_flip)^8 approximation	Verified empirically at 0.8580 across 30 independent trials	Mapped into decision controller thresholds.

4. Compliance Declaration

[X] No Internal Inspection: Decisions rely strictly on measurement counts returned by run(). Zero inspect or seed lookups.

[X] Import Isolation: solver.py imports only stdlib, numpy, scipy, qiskit. Never imports from local/.

[X] Clean Directory Execution: solver.py verified to execute standalone in an isolated directory (TEST-021).

[X] Query Budget Enforced: Charge-before-call QueryLedger guarantees query cost <= 60 under all branches.

Declaration: We confirm that this AI-use disclosure accurately reflects all generative AI contributions.